

Cluster-Based Pairs Trading: Combining Unsupervised Learning with Cointegration Filtering

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Abstract. We develop a market-neutral pairs trading strategy on a universe of 304 highly liquid exchange-traded funds (ETFs). Our motivation is to test whether statistically robust, market-neutral performance can be achieved under realistic constraints on data availability and computation. Daily adjusted prices are obtained from 2000 onward and restricted to ETFs with a common history from 6 November 2013 to 10 November 2025. Returns are standardized cross-sectionally by date and decomposed via principal component analysis; we consider both specifications that keep the first principal component (PC1) and specifications that remove PC1 to strip out the broad “market mode.” Using density-based clustering (HDBSCAN and OPTICS) on these standardized returns, we form ETF neighborhoods and, within each cluster, select candidate spreads via Engle–Granger cointegration together with a set of mean-reversion diagnostics. A rolling backtest with 12-month formation windows and 6-month trading windows trades all surviving spreads in an equally weighted, dollar-neutral portfolio using simple z -score entry/exit rules. Over the 16 December 2014 to 24 November 2025 trading period, our highest risk-adjusted performance comes from OPTICS clustering on PC1-removed returns, which delivers a cumulative return of 963.78%, an annualized Sharpe ratio of 2.4935, and a maximum drawdown of -8.74% (win rate 44.21%).

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1. Introduction

Modern pairs trading traces back to the quantitative equity group led by Nunzio Tartaglia at Morgan Stanley in the mid-1980s. Tartaglia’s team reportedly identified stocks whose prices tended to move together and designed rules to trade temporary divergences, generating substantial profits even through the 1987 crash (Gatev et al. 1999). The approach they developed—now widely referred to as *pairs trading*—has since become a standard relative-value technique in quantitative finance.

Pairs trading is a *market-neutral* strategy: instead of taking a directional bet on the overall market, the trader takes offsetting long and short positions in two related assets, typically with roughly equal dollar exposure, so that the net portfolio is largely hedged against broad market movements (Chen 2025). Here, a *long* position means buying an asset to profit if its price rises, while a *short* position means borrowing and selling an asset with the intent to buy it back later, profiting if its price falls. The central object is the *spread*, which we can think of as a single summary series that tracks how far the two assets have diverged from their usual relationship. If the spread is *mean-reverting*, unusually large divergences tend to shrink over time. In practice, we monitor a standardized version of the spread (a *z-score*) and enter trades only when the deviation is large: for example, enter when $|z_t|$ exceeds a threshold and exit once $|z_t|$ falls back toward a smaller band, with an explicit stop-loss if the deviation grows too extreme. Figure 1 illustrates these mechanics using an actual pair from our ETF backtest.

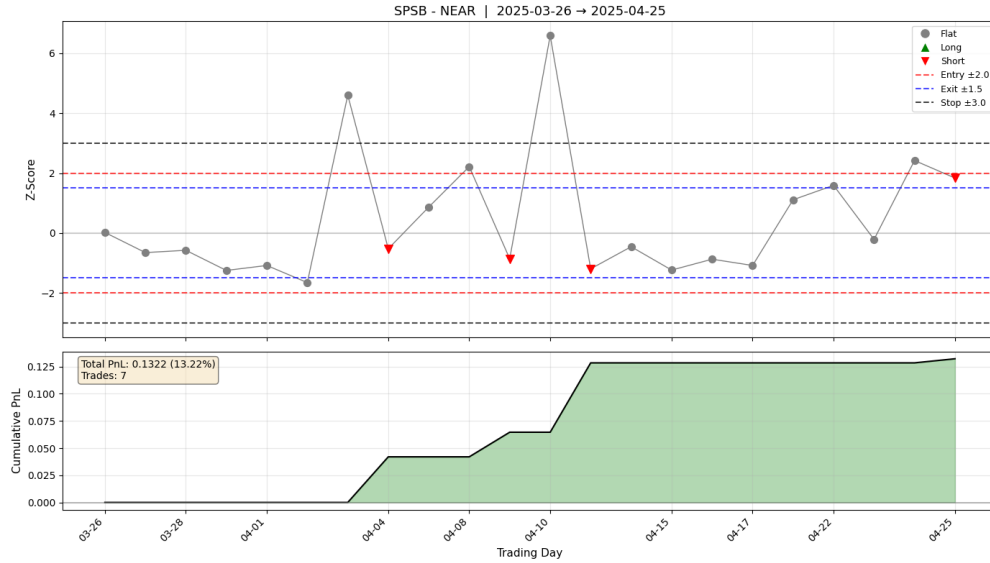


Figure 1: Pairs trading example from our backtest data: SPSB–NEAR over 2025-03-26 to 2025-04-25. **Top:** the spread z -score used for signal generation. Red dashed lines mark the entry thresholds at ± 2.0 , blue dashed lines mark the exit thresholds at ± 1.5 , and gray dashed lines mark the stop-loss thresholds at ± 3.0 . Markers indicate the position state on each day (flat, long, or short). **Bottom:** cumulative pair-level P&L (Profit and Loss) over the same window, with the total P&L and number of trades reported in the annotation.

Empirically, the canonical equity pairs trading study was [Gatev et al. \(1999, 2006\)](#). In their NBER working paper and subsequent *Review of Financial Studies* article in 2006, they worked with US stocks from 1967–1997 and adopted a rolling design with a 12-month *formation period* followed by a 6-month *trading period*. In each formation window, they normalized the price path of each stock by dividing by its initial price and, for each stock, selected the partner whose normalized path minimized the sum of squared daily deviations (a simple "distance" measure). They then kept the best-matched pairs and, in the trading window, monitored the difference between their normalized prices. A trade was opened when that difference exceeded a fixed threshold equal to two times its standard deviation estimated in the formation period; the strategy went long the recent underperformer and short the recent outperformer, and closed the position when the spread crossed back toward zero or the 6-month trading horizon ended. Even after conservative assumptions about transaction costs and short-sale fees, they reported economically meaningful excess returns in the earlier part of their sample, which helped establish pairs trading as a viable form of statistical arbitrage.

Subsequent work has focused on two main questions: how to select pairs and how to model the spread. On the selection side, [Rad et al. \(2016\)](#) compared three families of methods: (i) classical distance rules in the spirit of [Gatev et al. \(2006\)](#), (ii) cointegration-based methods, which explicitly tested whether a linear combination of two price series was statistically *stationary* (mean-reverting), and (iii) copula-based dependence models. They found that cointegration and copula methods often delivered higher risk-adjusted returns than pure distance rules, in part because they were more selective and focused on pairs whose spreads genuinely mean-reverted. This motivated the Engle–Granger cointegration filter ([Engle and Granger 1987](#)) we adopted in our ETF setting.

On the modeling side, it is common to approximate the spread as an *Ornstein–Uhlenbeck* (OU) process, a continuous-time mean-reverting diffusion. Under this assumption, the spread is characterized by a long-run mean, a speed of mean reversion, and a volatility parameter, which together determine quantities such as the expected *half-life* of a shock. [Zeng and Lee](#)

(2014) treated the usual entry and exit bands in a pairs strategy (for example, enter when the spread is more than two standard deviations from its mean and exit when it returns to the mean) as optimization variables. Given OU parameters and transaction costs, they show that the standard “rule-of-thumb” thresholds used in many empirical papers are not generally optimal, and that mis-specified thresholds can meaningfully reduce profitability. Related work by [Baviera and Santagostino Baldi \(2019\)](#) studies optimal leverage and stop-loss policies for mean-reverting statistical-arbitrage strategies under risk constraints and funding costs, showing that moderate leverage combined with explicit loss limits can raise the *Sharpe ratio* (mean excess return divided by return volatility) while keeping drawdowns small. These results support our emphasis on spreads with empirically estimated half-lives in a reasonable range and on using relatively conservative entry/exit bands in our backtest.

A more recent strand of the literature overlays machine learning on top of these classical designs. [Sarmiento and Horta \(2020a\)](#) and [Sarmiento and Horta \(2020b\)](#) keep distance-based or cointegration-based pairs selection but train predictive models on features of the spread and market environment to decide when to trade. They report that machine-learning timing rules can improve Sharpe ratios and reduce drawdowns relative to fixed threshold rules, while also highlighting overfitting risks and regime sensitivity. In this project, we deliberately stay on the simpler side—using rule-based z-score thresholds once we have identified candidate spreads—and focus instead on improving the quality of pair selection upstream. This emphasis naturally raises a practical question: under realistic limits on data access and computation, how much of the statistical-arbitrage performance documented in the literature can a fully reproducible, end-to-end pipeline recover?

1.1. Research Objective

Implementing pairs trading in practice depends not only on the trading rule, but also on the data and computation required to search for candidate pairs and validate stable spread dynamics. While the classical framework is well studied in U.S. equities, most published implementations either assume access to large cross-sectional universes, long CRSP-style histories, or even intraday data and sizable computing budgets. In practice, those assumptions are closer to the world of institutional statistical-arbitrage desks than to the constraints faced by an individual or small team working with free data and a single machine.

In this project we deliberately adopt the latter perspective and treat those constraints as design features. Instead of scanning thousands of individual stocks or intraday quotes, we work with a universe of 304 highly liquid ETFs whose daily adjusted prices are freely available from public sources. We restrict ourselves to end-of-day data (because that is what

a typical retail trader or student can obtain at scale at zero cost) and to methods that can be implemented with open-source Python libraries on a standard laptop. The question we ask is not “what would the best possible institutional pairs strategy look like?”, but:

How far can a carefully designed, fully reproducible pipeline built on free daily ETF data and modest compute go toward recovering the kind of statistical arbitrage opportunities documented in the literature?

More concretely, we organize our research around three questions:

1. **Can unsupervised learning on daily ETF returns, under realistic computational limits, recover economically meaningful asset groupings without relying on sector labels or fundamentals?** We apply a scalable unsupervised clustering pipeline to daily ETF returns to form data-driven ETF groupings.
2. **Within these clusters, can we identify pairs whose relationships exhibit persistent mean reversion rather than incidental short-run co-movement, using only daily data?** We filter candidate pairs using Engle–Granger cointegration and a small set of mean-reversion diagnostics computed from daily closes.
3. **Given these constraints, does a simple, transparent trading rule on the filtered spreads still deliver robust risk-adjusted performance?** We evaluate a rolling backtest of an equally weighted, dollar-neutral portfolio that trades the surviving spreads using simple z -score entry and exit rules, and we summarize performance via standard risk and turnover metrics.

Our contribution is therefore not a new theoretical model, but an *end-to-end empirical pipeline* that is intentionally constrained to what a well-motivated practitioner with free data and moderate computing power can actually run. We build a realistic ETF universe, use unsupervised clustering to avoid brute-force pair searches, filter pairs using cointegration and mean-reversion diagnostics calibrated to an Ornstein–Uhlenbeck-style model, and evaluate trading performance over rolling windows. We also compare variants that retain the market mode in clustering, or that replace HDBSCAN with OPTICS, to assess how sensitive the results are to the choice of clustering method under this “minimal resources” design.

The remainder of the paper is organized as follows. In Section 2, we describe our ETF universe, data collection, and preprocessing. Section 3 details the statistical procedures used for clustering, pair selection, and spread modelling. In Section 4, we present empirical results from both clustering and trading simulations. Finally, Section 5 concludes with interpre-

tation, limitations, and implications for real-world implementation in resource-constrained settings.

2. Data

2.1. Universe and motivation for using ETFs

Our analysis begins with a broad list of exchange-traded funds (ETFs)—pooled vehicles that hold baskets of assets and trade intraday on exchanges. We restrict to *non-leveraged* ETFs, meaning funds that do not use borrowing or derivatives to target a multiple of an index’s daily return, and *non-inverse* ETFs, meaning funds that do not aim to deliver the opposite (negative) of an index’s daily return. We impose these restrictions so that daily returns reflect underlying market movements without leverage-reset effects. We also exclude *exchange-traded notes (ETNs)*, which are unsecured debt securities issued by a financial institution that promise to pay the return of a reference index; unlike ETFs, ETNs are not asset-backed and therefore embed issuer credit risk.

ETFs are well suited for a pairs trading study because they smooth firm-specific earnings and event risk, producing cleaner estimates of cross-sectional co-movement from daily data. They also tend to be liquid, typically exhibiting high average *dollar volume* (price $\times$ traded shares) and relatively tight *bid-ask spreads* (the difference between the best quoted buy and sell prices). This reduces implementation frictions and supports short-selling, which is necessary for market-neutral spreads.

We begin with **414 unique tickers**, where a *ticker* is the short exchange-assigned symbol used to identify and trade a specific ETF (e.g., SPY or TLT), drawn to represent major risk dimensions across equities, fixed income, real estate, commodities, currencies, and thematic exposures. The complete list, along with the assignments of the broad categories, is reported in Appendix A. Only a subset of these tickers ultimately enter the empirical analysis, following the data-quality and liquidity screening described below.

2.2. Source, frequency, and return construction

We obtain end-of-day prices from Yahoo Finance using the `yfinance` Python API (Aroussi 2025; Yahoo Finance 2025). For each ETF i and trading day t , we keep the *adjusted close* price $P_{i,t}^{\text{adj}}$, which retroactively accounts for stock splits and cash distributions so that simple returns are economically meaningful.

Data are stored in a long panel with one row per (date, ticker). The simple daily percent

return for asset i on day t is

$$r_{i,t} = \frac{P_{i,t}^{\text{adj}}}{P_{i,t-1}^{\text{adj}}} - 1,$$

whenever both prices are observed. We do *not* forward-fill prices across missing days; *forward filling* means imputing a missing price by carrying forward the most recently observed value. If either side of the ratio is missing, the return for that date-ticker combination is dropped. The resulting tidy panel has four core columns:

$$\{\text{date}, \text{ticker}, \text{price} = P^{\text{adj}}, \text{pct_change} = r_{i,t}\}.$$

Any per-date or per-ticker standardization (for example, cross-sectional z -scores) is applied later as part of the clustering methodology and does not alter these raw inputs.

2.3. Quality and liquidity checks

Starting from the 414 tickers, we apply a sequence of rule-based *quality* and *liquidity* filters to obtain a universe that is both statistically reliable and realistically tradable:

- (i) **History depth.** Each ETF must have at least **3,000 trading days** of adjusted-close history in our raw download (roughly corresponding to an inception date no later than 2012). This provides enough depth for stable summary statistics and overlapping rolling windows.
- (ii) **Missingness.** Over the retained history, the daily return series must contain no missing values. If a ticker has gaps in adjusted closes that would break the return chain, it is dropped from the universe.
- (iii) **Volume coverage.** For each ETF, the fraction of trading days with reported volume equal to zero must be less than 1%. This rules out stale or effectively untradeable funds.
- (iv) **Median dollar volume.** We compute the median of Close $\times$ Volume over the active window and require it to be at least \$1,000,000. This serves as a coarse proxy for *liquidity*, i.e., the ability to transact near the quoted mid-price without large price impact.

After these filters, our final analysis universe consists of **304** highly liquid ETFs. In practice, every ETF that failed screening was removed by the 3,000-day history-depth requirement; the subsequent missing-data and liquidity checks served mainly to verify that the surviving 304 funds are clean and tradable.

2.4. Common history and sample period

Because pairs trading requires overlapping price histories, we next intersect the retained tickers over calendar time. The common *daily* adjusted–close history across all 304 ETFs runs from **16 December 2013** through **24 November 2025**, providing roughly 12 years of daily observations. The first twelve months of this span serve as the initial formation window for our rolling backtest.

For reporting and for the performance statistics in Section 4, we focus on the **16 December 2014** to **24 November 2025** trading span, which comprises **2,882 trading days** of *daily* adjusted closes assembled from Yahoo Finance. All subsequent clustering, cointegration tests, and trading simulations are based on this daily panel.

3. Methods

3.1. Overview

Our objective is to construct a fully reproducible workflow that identifies exchange–traded fund (ETF) pairs exhibiting persistent mean reversion and evaluates whether such pairs produce robust trading performance over time. The pipeline operates on daily adjusted closing prices and consists of four sequential steps:

1. Dimensionality reduction of standardized return series.
2. Unsupervised clustering to form economically coherent candidate neighborhoods.
3. Statistical pair selection via cointegration and mean–reversion diagnostics.
4. Rolling-window trading simulation with out-of-sample evaluation.

Steps 1–2 use only cross-sectionally standardized daily returns to extract a low-dimensional structure and to form candidate ETF neighborhoods via density-based clustering. Step 3 operates within these neighborhoods to select spreads with strong empirical evidence of mean reversion. Step 4 implements a dollar-neutral trading strategy on the selected spreads in rolling formation and trading windows, meaning the long and short legs are sized to have equal dollar exposure (so the net market value of the position is approximately zero), as described in later subsections.

3.2. Step 1: Dimensionality reduction of standardized returns

Let $r_{i,t}$ denote the simple daily return of ETF i on trading day t , computed from adjusted closes as in Section 2.2. For a given formation window with trading days $t = 1, \dots, T$ and ETFs $i = 1, \dots, N$, we start from the $T \times N$ return matrix

$$R = (r_{i,t})_{t=1,\dots,T; i=1,\dots,N}.$$

Cross-sectional standardization. Consistent with the design stated in our first research question (Section 1.1), we standardize returns *cross-sectionally by date*. For each day t , let

$$\bar{r}_t = \frac{1}{N} \sum_{i=1}^N r_{i,t}, \quad s_t^2 = \frac{1}{N-1} \sum_{i=1}^N (r_{i,t} - \bar{r}_t)^2,$$

denote the cross-sectional mean and variance of ETF returns at date t . We define *standardized returns*

$$x_{i,t} = \frac{r_{i,t} - \bar{r}_t}{s_t}.$$

In matrix form, this yields a standardized panel

$$X = (x_{i,t})_{t=1,\dots,T; i=1,\dots,N},$$

in which each date t has mean zero and unit variance across ETFs. In code, this corresponds to pivoting returns into a ticker-by-date matrix and applying `StandardScaler` (from `scikit-learn`) across columns, so that each date column has mean zero and unit variance across tickers (Pedregosa et al. 2011).

Principal component analysis and market mode. We apply principal component analysis (PCA) to the standardized returns X to extract dominant co-movement patterns (Jolliffe and Cadima 2016; Shlens 2014). Let

$$X = U\Sigma V^\top$$

denote the singular value decomposition, where the columns of V are orthonormal loading vectors (“principal directions”) ordered by explained variance. The first principal component (PC1) typically corresponds to a broad “market mode” in which most ETFs move together (Plerou et al. 2002).

We consider two representations:

- **PC1-kept (raw residual structure).** We retain the standardized return matrix X as is and measure distances directly in this space.

- **PC1-removed (market-neutral residuals).** We remove the market mode by projecting out PC1. Let v_1 denote the first loading vector and $f_1 = Xv_1$ the associated PC1 time series. Here, $f_1 \in \mathbb{R}^T$ is the *common daily factor*: for each day t , $(f_1)_t$ is the portfolio return obtained by weighting ETFs by the PC1 loadings in v_1 , and it captures the dominant cross-sectional co-movement shared by most ETFs. We form residualized returns

$$X^{(-1)} = X - f_1 v_1^\top,$$

which removes the rank-one market component and emphasizes sector- and style-specific co-movement.

Unless otherwise stated, we use $X^{(-1)}$ when we speak of “PC1-removed” returns and X when PC1 is “kept.”

Correlation distance. For clustering we work with a correlation-based distance between ETFs. Given two ETFs i and j , let ρ_{ij} denote the Pearson correlation between their standardized return vectors (either from X or $X^{(-1)}$ over the formation window). We define the *correlation distance* as:

$$d_{\text{corr}}(i, j) = 1 - \rho_{ij}.$$

This lies in $[0, 2]$, equals zero when two series are perfectly positively correlated, and increases as they become less correlated or negatively correlated. In our implementation this is computed via `pairwise_distances` with `metric="correlation"`, which returns precisely $1 - \rho_{ij}$ for each pair. The resulting distance matrix $D = (d_{\text{corr}}(i, j))$ serves as the input to both HDBSCAN and OPTICS in the next step.

3.3. Step 2: Density-based clustering

After dimensionality reduction and standardization, we cluster ETFs using density-based methods applied to the correlation-distance matrix D . The goal is to identify neighborhoods of ETFs that exhibit similar residual return dynamics—i.e., comparable co-movement patterns after removing broad market effects—so that cointegration tests and mean-reversion diagnostics are applied primarily within economically coherent groups, reducing both multiple-testing risk and computational burden relative to enumerating all $\binom{N}{2}$ pairs.

We use two complementary algorithms:

- (i) **HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise)**, which builds a hierarchy of density clusters and extracts those that remain stable as the density threshold varies ([Campello et al. 2013](#); [McInnes et al. 2017](#)).

- (ii) **OPTICS (Ordering Points To Identify the Clustering Structure)**, which orders points to reveal multi-scale “valleys” of high density without committing to a single global density cutoff (Ankerst et al. 1999).

Both methods treat ETFs that do not belong to any sufficiently dense group as *noise*, which in our context corresponds to funds whose standardized returns do not clearly belong to any co-moving family.

HDBSCAN. HDBSCAN starts from a notion of *core distance*, the distance from a point to its `min_samples`-th nearest neighbor. Using the correlation distance d_{corr} , the core distance of ETF i is the k -th order statistic of $\{d_{\text{corr}}(i, j)\}_{j \neq i}$ for $k = \text{min_samples}$. The *mutual reachability distance* between two ETFs a and b is defined as

$$d_{\text{mreach}}(a, b) = \max\{d_{\text{core}}(a), d_{\text{core}}(b), d_{\text{corr}}(a, b)\}.$$

HDBSCAN constructs a hierarchy by viewing density through the mutual-reachability metric. After computing the mutual-reachability graph, it builds a minimum-spanning tree (MST) whose edge weights define the critical distance scales. Writing $\lambda = 1/d_{\text{mreach}}$ (so larger λ corresponds to a tighter, higher-density scale), the algorithm forms a nested sequence of clusters by sweeping λ upward and tracking the connected components that remain as edges with large d_{mreach} are removed. This yields a single-linkage style cluster tree. To obtain a flat partition, HDBSCAN then *condenses* the tree by pruning branches that fall below `min_cluster_size` and selects the set of clusters that maximizes a stability criterion (clusters that persist across a wide range of λ receive higher stability). Points not assigned to any selected stable cluster are labeled as noise.

In our implementation we run HDBSCAN with `min_cluster_size` = 3 and `min_samples` = 2 on two representations:

- the PC1-kept standardized returns X ; and
- the PC1-removed residuals $X^{(-1)}$.

Here, `min_cluster_size` sets the smallest group of ETFs that will be retained as a cluster in the condensed tree (smaller branches are pruned and treated as noise), while `min_samples` controls how conservatively HDBSCAN defines local density via the core distance (larger values require more nearby neighbors for a point to be considered in a dense region and typically increase the amount of noise). In both cases we supply the precomputed correlation-distance matrix D and let HDBSCAN infer cluster shapes without assuming convexity or a fixed number of clusters.

Figure 2 shows the condensed cluster tree for the PC1-removed specification; long high- λ branches correspond to density-stable ETF clusters that HDBSCAN retains, while short branches tend to be discarded as transient groupings.

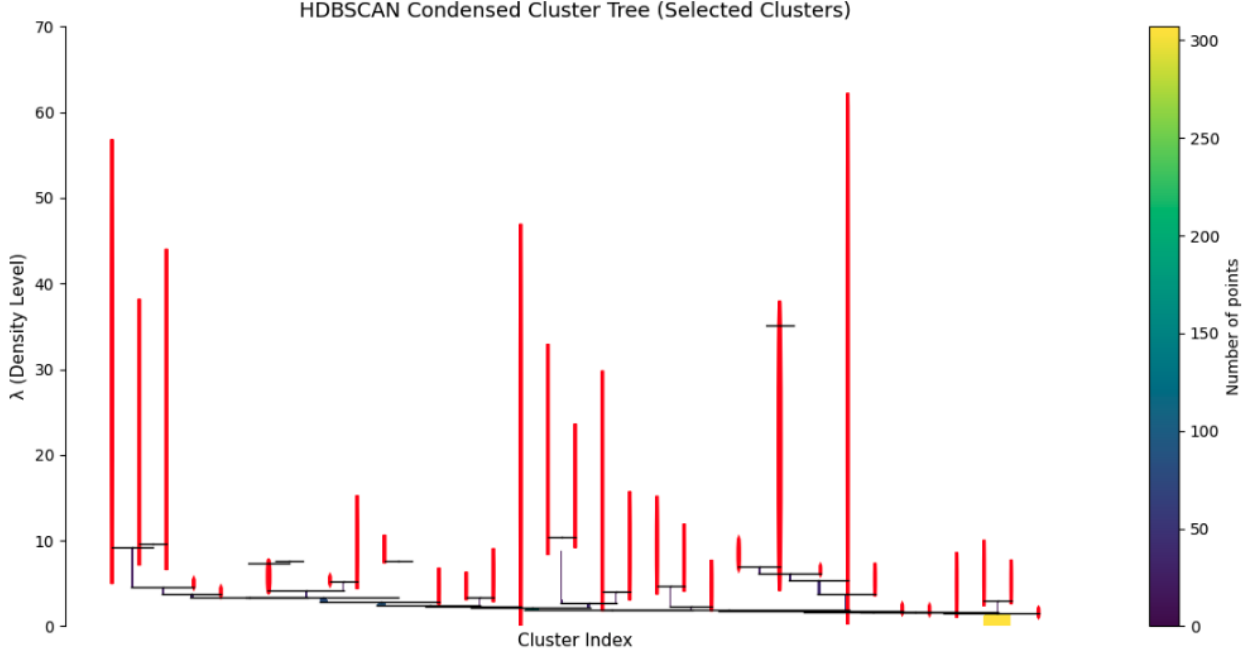


Figure 2: HDBSCAN condensed cluster tree on the PC1-removed correlation-distance matrix. Long high- λ branches indicate density-stable ETF clusters retained by the algorithm; points not belonging to any stable branch are labeled as noise.

OPTICS. OPTICS constructs a multi-scale density view of the data by producing (i) an ordering of ETFs and (ii) a *reachability distance* value for each ETF in that ordering (Ankerst et al. 1999), rather than committing to a single global density threshold. Fix `min_samples` = k and a metric d (here the correlation distance). For each ETF p , define its *core distance*

$$d_{\text{core}}(p) = \text{the } k\text{-th nearest-neighbor distance of } p \text{ under } d,$$

with the convention $d_{\text{core}}(p) = \infty$ if fewer than k neighbors exist. If o is a previously processed ETF, the *reachability distance* of a candidate ETF p from o is

$$d_{\text{reach}}(p \mid o) = \max\{d_{\text{core}}(o), d(o, p)\}.$$

OPTICS iteratively selects the next ETF with the smallest current reachability distance (maintained via a priority queue) and updates the reachability distances of its neighbors.

Dense regions appear as *valleys* of low reachability in the resulting profile, while transitions between groups appear as ridges.

We run OPTICS with `min_samples= 2`, `min_cluster_size= 2`, and $\xi = 0.1$ on two representations:

- the PC1-kept standardized returns X ; and
- the PC1-removed residuals $X^{(-1)}$.

Here, `min_samples` sets the neighborhood size used to define local density (and thus the core distance), `min_cluster_size` is the smallest group the extraction routine will return as a cluster, and ξ is the reachability “steepness” threshold that determines how sharp an upward transition must be to mark a cluster boundary (larger ξ tends to yield more, smaller micro-clusters). In both cases we supply the corresponding precomputed correlation-distance matrix D and extract clusters from the reachability plot using the ξ -steepness criterion, which identifies cluster boundaries via sufficiently sharp upward transitions in reachability and enforces a minimum cluster size.

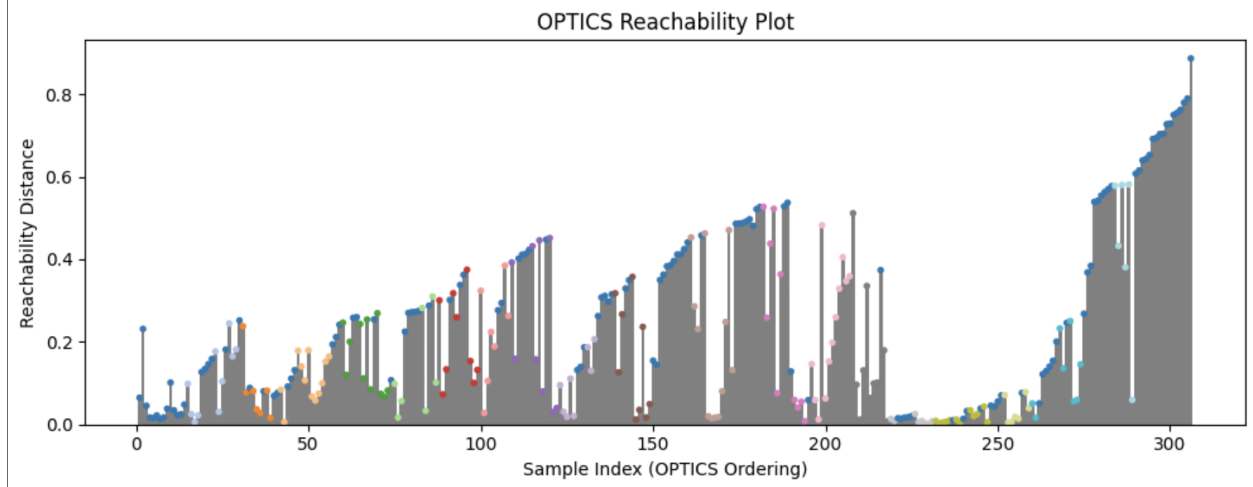


Figure 3: OPTICS reachability plot on the PC1-kept correlation-distance matrix. Valleys (low reachability) correspond to dense ETF micro-clusters; ridges separate clusters. Colors indicate the final OPTICS cluster assignments, with gray points typically denoting noise.

Clustering summary. Table 1 summarizes the main clustering outcomes on the $N = 304$ -ETF universe. For HDBSCAN we report both the PC1-kept and PC1-removed specifications; for OPTICS we report the PC1-kept specification used to generate the reachability plot.

Table 1: Clustering comparison on standardized ETF return matrix ($N = 304$). Noise = tickers not assigned to multi-member clusters.

Method	PC1	Parameters	Clusters	Noise	Noise%	Comment
HDBSCAN	✗	mcs = 3 ms = 2	31	143	47.0	Market neutral
HDBSCAN	✓	mcs = 3 ms = 2	31	143	47.0	No assumption
OPTICS	✗	mcs = 2 ms = 2 $\xi = 0.1$	65	107	35.2	Market neutral, small clusters
OPTICS	✓	mcs = 2 ms = 2 $\xi = 0.1$	64	114	37.5	Small clusters

Note: `mcs` denotes `min_cluster_size` (the minimum number of ETFs required for a group to be retained as a cluster), `ms` denotes `min_samples` (the neighborhood size used to define core distance/local density), and ξ is the OPTICS steepness threshold used to extract clusters from the reachability plot.

Both algorithms identify economically interpretable clusters: for example, broad U.S. equity index trackers (e.g., IVV, VTI, SCHB, ITOT), large-cap growth funds (e.g., IWF, VUG, SCHG), sector funds (e.g., XLF/Ryf/IYF for financials; XLK/VGT/FTEC for technology), and commodity or rate complexes (e.g., oil and agriculture ETFs; Treasury duration buckets). OPTICS tends to produce a larger number of small clusters, which is useful for surfacing tight “clone” relationships, while HDBSCAN emphasizes more persistent, density-stable groups. In the remainder of the paper we treat these clusters as candidate neighborhoods and restrict all subsequent pair selection and mean-reversion diagnostics to ETF pairs within the same cluster.

3.4. Step 3: Pair formation and testing

Given a clustering of ETFs in return space, we next search for statistically robust, economically interpretable spreads within each cluster. For every unordered pair (i, j) of tickers belonging to the same cluster, we estimate a candidate spread on log prices and subject it to a sequence of mean-reversion filters.

Engle–Granger cointegration (ADF test). Let $p_{i,t} = \log P_{i,t}$ denote the log adjusted close of ETF i on day t in a given formation window. For each ordered pair (i, j) within a cluster, we estimate the hedge ratio $\hat{\beta}_{ij}$ and intercept $\hat{\alpha}_{ij}$ from the static regression

$$p_{i,t} = \alpha_{ij} + \beta_{ij}p_{j,t} + u_{ij,t},$$

and define the spread regression residual as:

$$s_{ij,t} = p_{i,t} - \hat{\beta}_{ij} p_{j,t} - \hat{\alpha}_{ij}.$$

We then apply an ADF unit-root test to $s_{ij,t}$ (equivalently, to the demeaned residuals $\tilde{s}_{ij,t}$). The ADF test fits an autoregression in first differences with lagged differences as controls and tests whether the lagged level term is zero (unit root) versus negative (stationary), so rejecting the null indicates that the spread is stationary and hence consistent with mean reversion. Following the standard Engle–Granger two-step procedure (Engle and Granger 1987), a pair is retained only if we can reject the null of a unit root at the 5% level. In our ETF universe, cointegration is relatively rare: roughly 69% of within-cluster pairs fail the ADF test at this threshold and are discarded at this stage. The remaining pairs have spreads that are at least theoretically mean-reverting in the long run.

Hurst-exponent filter. Cointegration alone does not guarantee that a spread exhibits the kind of short- to medium-horizon mean reversion that is attractive for trading. To refine the set further, we compute a Hurst exponent H_{ij} for the spread $s_{ij,t}$ on the same formation window, using a rescaled-range (R/S) statistic that estimates the scaling law $\mathbb{E}[R(n)/S(n)] \propto n^H$, where H summarizes the series’ long-memory: $H < 0.5$ indicates anti-persistence (mean reversion), $H = 0.5$ corresponds to a random-walk-like process, and $H > 0.5$ indicates persistence (trending). Values $H < 0.5$ indicate anti-persistent / mean-reverting behavior, while $H > 0.5$ corresponds to persistent trending. Extremely low H close to zero typically reflect almost white-noise series with very little structure.

We therefore retain only spreads with

$$0.20 < H_{ij} < 0.50,$$

which filters out both strongly trending series and extremely noisy series that are unlikely to deliver exploitable excursions. In our baseline implementation, about 19.2% of ADF-cointegrated pairs fail this Hurst filter and are removed.

OU half-life filter. For the remaining spreads, we estimate an approximate Ornstein–Uhlenbeck (OU) mean-reversion speed by fitting an AR(1) model to the demeaned spread,

$$\Delta s_{ij,t} = \phi_{ij} s_{ij,t-1} + \eta_{ij,t},$$

where an AR(1) specification means that today’s change in the spread is modeled as a linear function of yesterday’s level plus a noise term; when $\hat{\phi}_{ij} < 0$, this implies mean reversion

toward the long-run mean. We then convert the AR(1) coefficient $\hat{\phi}_{ij}$ into an implied half-life (in trading days),

$$t_{ij}^{(1/2)} = -\frac{\ln(2)}{\hat{\phi}_{ij}}.$$

This is the horizon over which a typical shock to the spread decays to half its initial magnitude under the OU approximation (see, e.g., [Zeng and Lee 2014](#); [Rad et al. 2016](#)).

We keep only spreads with

$$3 \leq t_{ij}^{(1/2)} \leq 25 \quad \text{trading days.}$$

Very short half-lives imply extremely rapid mean reversion: with only one observation per trading day, such dynamics would be largely invisible in daily data, and the resulting excursions tend to be small and noisy once transaction costs are taken into account. Very long half-lives, by contrast, imply that a position may need to be held for months before converging, which is operationally unattractive in our design. Pairs with estimated half-life outside this band are discarded. In our data, this removes an additional 4.7% of spreads that had passed the ADF and Hurst filters.

Excursion requirement. Finally, even a statistically mean-reverting spread may not be practically tradeable if it almost never deviates from its mean. For each surviving spread, we compute a *formation-window z-score*:

$$z_{ij,t}^{\text{form}} = \frac{s_{ij,t} - \bar{s}_{ij}}{\hat{\sigma}_{ij}},$$

where $\bar{s}_{ij}$ and $\hat{\sigma}_{ij}$ are the sample mean and standard deviation of $s_{ij,t}$ over the formation window. We require that the spread exhibit a small but non-trivial number of excursions beyond ± 2 standard deviations in this period; pairs that never meaningfully leave the “no-trade” region are dropped since they are unlikely to yield actionable signals under the entry/exit rules described below. The surviving set of pairs constitutes the candidate spread universe for the subsequent trading window.

3.5. Pair Trade Signal and Implementation

For each selected pair (i, j) , we use the Engle–Granger regression ([Engle and Granger 1987](#)) from the formation window to define a fixed hedge ratio $\hat{\beta}_{ij}$ over the subsequent trading window. The trading spread at date t is

$$s_{ij,t} = p_{i,t} - \hat{\beta}_{ij} p_{j,t},$$

where $p_{i,t}$ and $p_{j,t}$ are log prices. We monitor this spread using a rolling z -score computed from recent trading-window history.

Rolling z -score and thresholds. On each trading day t , we compute a 20-day rolling mean and standard deviation for $s_{ij,t}$,

$$\mu_{ij,t}^{(20)} = \frac{1}{20} \sum_{k=0}^{19} s_{ij,t-k}, \quad \sigma_{ij,t}^{(20)} = \sqrt{\frac{1}{19} \sum_{k=0}^{19} (s_{ij,t-k} - \mu_{ij,t}^{(20)})^2},$$

and define the standardized spread

$$z_{ij,t} = \frac{s_{ij,t} - \mu_{ij,t}^{(20)}}{\sigma_{ij,t}^{(20)}}.$$

Trading signals follow a simple two-threshold rule, in the spirit of [Gatev et al. \(2006\)](#) and OU-based optimizations such as [Zeng and Lee \(2014\)](#):

- **Entry:** initiate a position when $|z_{ij,t}| \geq 2$ and no trade in (i, j) is currently open.
 - If $z_{ij,t} \geq 2$, the spread is “rich”: we short the spread (short i , long j).
 - If $z_{ij,t} \leq -2$, the spread is “cheap”: we go long the spread (long i , short j).
- **Exit:** close the position when $|z_{ij,t}| \leq 1$ or the trading window ends, whichever comes first.

The exit threshold is set strictly inside the entry band rather than at $z_{ij,t} = 0$. In recent ETF markets, spreads frequently converge partway toward equilibrium without fully returning to their long-run mean; a tighter exit captures this partial reversion and avoids extended holding risk with diminishing incremental profit ([Song and Zhang 2013](#); [Zeng and Lee 2014](#)).

Self-financing, dollar-neutral execution. Positions are implemented with equal notional size on the long and short legs, scaled by the hedge ratio. On entry for pair (i, j) , we choose portfolio weights $(w_{i,t}, w_{j,t})$ such that

$$w_{i,t} P_{i,t} = \hat{\beta}_{ij} w_{j,t} P_{j,t},$$

so that the long and short legs have equal dollar magnitude and the net position is approximately market-neutral. The trade is *self-financing*: the proceeds from the short leg finance the long leg, and we do not apply additional leverage beyond this dollar-neutral construction.

Daily P&L for pair (i, j) is then computed from price changes on both legs,

$$\Delta \Pi_{ij,t+1} = w_{i,t} (P_{i,t+1} - P_{i,t}) + w_{j,t} (P_{j,t+1} - P_{j,t}),$$

accumulated over the holding period from entry to exit. At the portfolio level, we sum P&L across all active pairs and maintain equal weight across open spreads (subject to the constraint that each individual spread is itself dollar-neutral).

3.6. Step 4: Trading simulation

We evaluate the full strategy using a *rolling formation/trading design* (Gatev et al. 2006). Windows overlap so that clusters, spreads, and trading signals can adapt to evolving return dynamics while preserving a strictly out-of-sample evaluation of performance.

Window structure. The timeline is partitioned into a sequence of non-overlapping formation windows and subsequent trading windows:

- **Formation window:** 12 months of daily data, used for clustering, pair selection, and estimation of $\hat{\beta}_{ij}$, H_{ij} , and $t_{ij}^{(1/2)}$.
- **Trading window:** the 6 months immediately following each formation window, during which the entry/exit rules in Section 3.5 are applied to the fixed list of pairs selected from that formation period.

We roll this 12+6 month cycle forward through the sample. At any calendar date, the portfolio may contain positions initiated under different formation windows, but each individual trade is always generated using information strictly prior to its own trading window.

Backtest structure. Algorithm 1 summarizes the construction of candidate spreads within a single formation window, combining clustering (Section 3.3) with the cointegration and mean-reversion filters of Section 3.4. Algorithm 0 then describes the rolling backtest that trades these spreads using the z -score rules and self-financing execution in Section 3.5.

We apply Algorithms 1 and 0 separately under each clustering specification described in Section 3.3: HDBSCAN on PC1-removed residual returns, HDBSCAN on PC1-kept standardized returns, and OPTICS on PC1-kept standardized returns. In every case, the clustering step only changes the set of candidate within-cluster pairs; all subsequent filters, trading rules, and portfolio construction are held fixed.

Algorithm 1 Pair selection in formation window k

- 1: **Input:** daily adjusted closes $P_{i,t}$ for all ETFs $i = 1, \dots, N$ and dates $t \in \mathcal{T}_k^{\text{form}}$.
- 2: Compute simple daily returns $r_{i,t}$ and cross-sectionally standardized returns $x_{i,t}$; form panel X .

end for

12: end for

Aggregate P&L across all windows and pairs; compute portfolio statistics (cumulative return, annualized Sharpe, max drawdown, turnover, etc.).

4. Results

We evaluate the rolling pairs trading strategy described in Section 3 over the 2014–2025 sample on the $N = 304$ ETF universe. Table 2 summarizes full-sample performance for all four clustering specifications: OPTICS with PC1 removed, OPTICS with PC1 kept, HDBSCAN with PC1 removed, and HDBSCAN with PC1 kept. Figures 4–7 show the corresponding cumulative return paths.

Clustering	PC1 Removal	Cumulative Return	Sharpe Ratio	Max Drawdown	Win Rate
OPTICS	×	449.48%	1.7228	-12.71%	45.18%
OPTICS	✓	963.78%	2.4935	-8.74%	44.21%
HDBSCAN	×	612.86%	2.0087	-18.79%	46.32%
HDBSCAN	✓	1020.34%	2.0755	-28.39%	44.31%
S&P 500	N/A	302.03%	0.8041	-33.72%	N/A
Dow Jones	N/A	170.93%	0.6097	-37.09%	N/A
Nasdaq	N/A	542.53%	0.8847	-35.12%	N/A

Table 2: Backtesting Results from 2014-12-16—2025-11-24

4.1. Metrics and overall performance

Table 2 reports four summary statistics for each specification. Let r_t denote the strategy’s daily portfolio return and let V_t denote the corresponding cumulative wealth index, initialized at $V_0 = 1$ and updated by

$$V_t = V_{t-1}(1 + r_t).$$

Cumulative Return is the total growth over the backtest,

$$R_{\text{cum}} = \frac{V_T}{V_0} - 1 = V_T - 1.$$

Sharpe Ratio is the annualized mean–variance ratio of daily returns,

$$\text{Sharpe} = \sqrt{252} \frac{\bar{r}}{s(r)},$$

where $\bar{r}$ and $s(r)$ are the sample mean and standard deviation of $\{r_t\}$.

Max Drawdown is the worst peak-to-trough decline of the wealth index,

$$\text{MDD} = \min_t \left(\frac{V_t}{\max_{u \leq t} V_u} - 1 \right).$$

Win Rate is the fraction of trading days with positive returns,

$$\text{WinRate} = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{r_t > 0\}.$$

Across the four clustering specifications, cumulative returns range from 449.48% (OPTICS, PC1 kept) to 1020.34% (HDBSCAN, PC1 removed), corresponding to terminal wealth multipliers of

$$1 + 4.4948 = 5.4948 \quad \text{to} \quad 1 + 10.2034 = 11.2034.$$

Sharpe ratios range from 1.7228 to 2.4935, with the highest Sharpe achieved by OPTICS with PC1 removed. Maximum drawdowns range from -12.71% (smallest drawdown) to -28.39% (largest drawdown). Win rates fall between 44.21% and 46.32%, indicating the strategy is profitable on fewer than half of days, but with sufficiently asymmetric gains versus losses to generate strong risk-adjusted performance.

4.2. Effect of clustering and market-mode removal

Table 2 shows that the impact of market-mode (PC1) removal depends on the clustering method. Under OPTICS, removing PC1 improves performance materially: cumulative return rises from 449.48% (PC1 kept) to 963.78% (PC1 removed) and the Sharpe ratio increases from 1.7228 to 2.4935, while maximum drawdown improves from -12.71% to -8.74% . This pattern is consistent with the interpretation that, for OPTICS, projecting out PC1 helps the distance geometry emphasize sector- and style-driven co-movement that is more relevant for identifying tradable mean-reverting spreads.

In contrast, under HDBSCAN, PC1 removal increases total return but worsens downside risk. Cumulative return increases from 612.86% (PC1 kept) to 1020.34% (PC1 removed), but the Sharpe ratio changes only modestly (from 2.0087 to 2.0755) and maximum drawdown deteriorates from -18.79% to -28.39% . Thus, while removing PC1 can expand the opportunity set and raise terminal wealth, it may also admit clusters (and hence pairs) that are more vulnerable to large adverse excursions.

Comparing clustering methods, OPTICS with PC1 removed achieves the strongest risk-adjusted performance (highest Sharpe and smallest drawdown), while HDBSCAN with PC1 removed achieves the highest terminal wealth but with the largest drawdown. Win rates

are broadly similar across specifications (roughly 44%–46%), suggesting that differences in overall performance are driven more by payoff asymmetry and tail behavior than by the fraction of winning days.

4.3. Equity curves

Figures 4–7 plot cumulative returns for the three specifications. All series display a broadly monotone growth profile with limited large drawdowns, consistent with the mean-reverting, dollar-neutral design of the strategy. Performance is relatively steady through different market regimes, including the 2015–2016 commodity downturn, the 2018 volatility spike, the 2020 COVID crash and rebound, and the 2022–2023 rate-hiking cycle, indicating that the pairs-selection filters are effective at avoiding structurally unstable spreads.

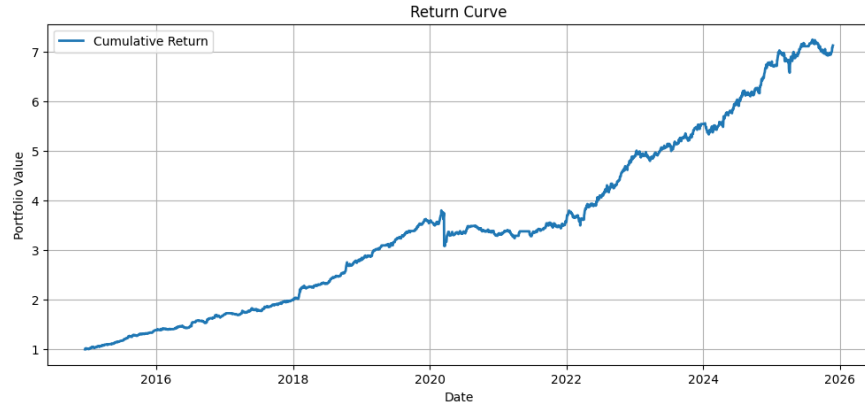


Figure 4: Cumulative return of the pairs trading strategy using HDBSCAN clusters (PC1 kept).

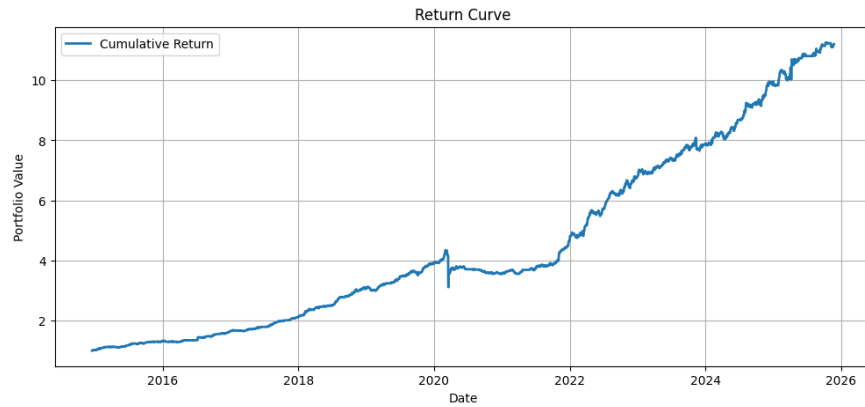


Figure 5: Cumulative return of the pairs trading strategy using HDBSCAN clusters (PC1 removed).

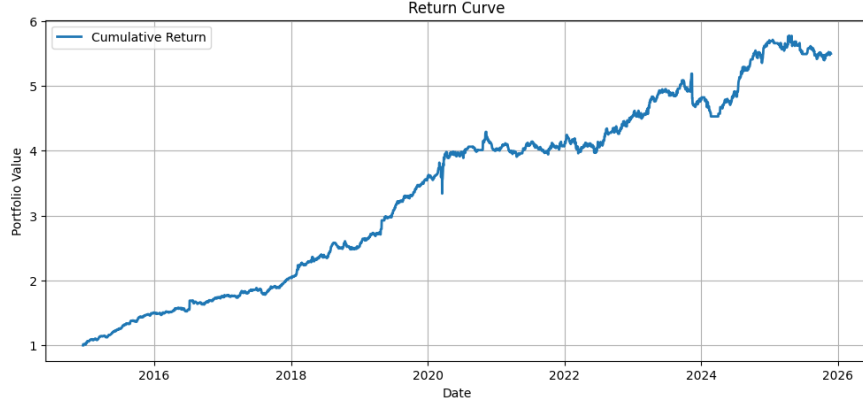


Figure 6: Cumulative return of the pairs trading strategy using OPTICS clusters (PC1 kept).

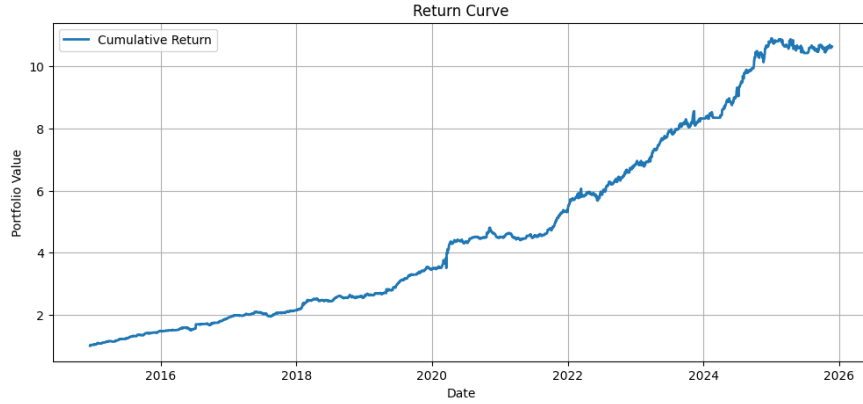


Figure 7: Cumulative return of the pairs trading strategy using OPTICS clusters (PC1 removed).

5. Discussion

The performance we obtain is encouraging given the constraints of our design: we use only daily adjusted closes, transparent mean-reversion filters, and simple z -score entry/exit rules without any intraday microstructure signals. Across the four clustering specifications in Table 2, annualized Sharpe ratios range from 1.7228 to 2.4935. The best risk-adjusted variant is OPTICS with PC1 removed (Sharpe 2.4935), which also achieves the smallest maximum drawdown (-8.74%). At the same time, the highest terminal wealth is achieved by HDBSCAN with PC1 removed (cumulative return 1020.34%), albeit with materially larger downside risk (max drawdown -28.39%). Overall, the results suggest that a fully reproducible, daily-data pipeline can still recover substantial relative-value structure in a broad ETF universe.

A natural benchmark is a passive investor who buys and holds a broad equity index over the same window. Over 2014–2025, Table 2 shows that SPY (S&P 500) delivers a cumulative return of 302.03% (wealth multiplier $1 + 3.0203 = 4.0203$) while QQQ (Nasdaq-100) delivers 542.53% (wealth multiplier $1 + 5.4253 = 6.4253$). By contrast, our pairs trading specifications produce cumulative returns between 449.48% and 1020.34%, corresponding to wealth multipliers between 5.4948 and 11.2034. Figure 8 makes this comparison concrete by plotting the growth of \$1 in our best-performing specification (HDBSCAN with PC1 removed) against both SPY and QQQ over the backtest window. Notably, the long-only benchmarks experience deep peak-to-trough losses (max drawdowns of -33.72% for SPY and -35.12% for QQQ in our sample), whereas our market-neutral portfolio exhibits a smaller drawdown (-28.39%), consistent with the strategy’s intent to reduce broad equity exposure.¹

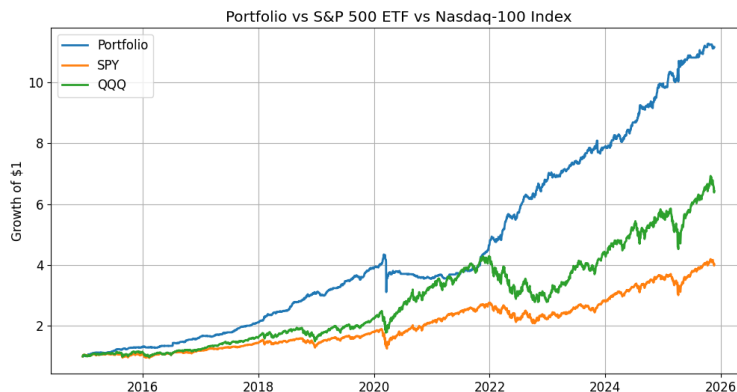


Figure 8: Cumulative total return of the pairs-trading portfolio compared with two long-only equity benchmarks (SPY and QQQ) over the backtest window (2014–2025). The plot shows the growth of \$1 invested in each series, where SPY tracks the S&P 500 and QQQ tracks the Nasdaq-100; both serve as benchmark references for our market-neutral strategy.

The comparison between PC1-kept and PC1-removed variants is also informative. Removing PC1 strips out the dominant cross-sectional “market mode” before distances are computed, which can sharpen sector- and style-driven similarity and potentially improve the quality of candidate neighborhoods. Empirically, the effect depends on the clustering method. Under OPTICS, removing PC1 improves performance on every metric we report: cumulative return rises from 449.48% to 963.78%, Sharpe increases from 1.7228 to 2.4935, and max drawdown improves from -12.71% to -8.74%. Under HDBSCAN, PC1 removal increases cumulative return (612.86% to 1020.34%) but substantially worsens tail risk (max drawdown -18.79%

¹Benchmark calculations use adjusted-close data from Yahoo Finance over the same 2014-12-16 to 2025-11-24 window as the strategy backtest.

to -28.39%), with only a modest Sharpe change (2.0087 to 2.0755). This contrast suggests that market-mode removal is not universally beneficial: it can either isolate cleaner residual co-movement that supports stable mean reversion, or it can admit clusters whose spreads are more vulnerable to large adverse excursions.

Our results also highlight the scarcity of spreads that remain tradeable after strict statistical filtering. Cointegration is demanding, and the subsequent diagnostics (Hurst, OU half-life, excursion requirement) remove a large fraction of within-cluster pairs. In early tests with smaller universes, this sometimes produced an unstable opportunity set with too few eligible spreads per window. Expanding to a 300+ ETF universe mitigates this and yields a more diversified portfolio, but the acceptance rate after all filters remains low. For a production implementation that targets broad diversification across sectors, geographies, and risk factors while keeping any single pair weight small, a substantially larger universe (e.g., thousands of single-name equities) would likely be required.

Finally, several practical considerations remain outside our backtest. Because the strategy is self-financing and requires short positions, borrow fees, ETF lending costs, and execution frictions would reduce live performance. Likewise, using daily closes means we do not model intraday volatility, limit-order execution, or slippage around market opens/closes. Even with these caveats, the evidence suggests that clustering-based neighborhood selection combined with strict mean-reversion diagnostics can extract a robust relative-value signal from daily ETF data over more than a decade of observations.

Appendix

A. Universe Construction Details

This appendix expands on the ETF selection described in Section 2. The full candidate universe consists of 414 unique ETF tickers drawn from major issuers (BlackRock iShares, Vanguard, State Street SPDR, Invesco) to ensure liquidity, availability of historical data, and stability of fund mandates. The list spans broad equity, factor portfolios, sectors, international regions, single-country funds, fixed income, real estate, commodities, and thematic exposures.

A.1. Category structure and full ticker list

The universe definition follows a structured block layout that groups related funds:

- **U.S. Equity (Broad / Style):** market-cap and style funds (e.g., IVV, VTI, IWF, IWD).
- **U.S. Sectors:** GICS-based sector funds (e.g., XLK, XLE, XLF) with equal-weight analogs (e.g., RYT).
- **U.S. Industry / Theme:** focused industry exposures (e.g., SMH, SOXX, IBB, XRT).
- **International (Broad / Regions):** developed and emerging market aggregates (e.g., ACWI, IEFA, VWO).
- **Developed Single-Country:** large-market country trackers (e.g., EWJ, EWU, EWG, EWA).
- **Emerging Single-Country:** liquid EM funds (e.g., EWZ, EWT, INDA, EZA, FM).
- **Currency-Hedged Variants:** e.g., HEFA, HEDJ, retained to allow hedged–unhedged comparisons.
- **Fixed Income (Treasury, Credit, TIPS, MBS, Munis, Preferreds, Loans):** BND, TLT, LQD, HYG, TIP, MBB, MUB, PFF, BKLN, etc.
- **Real Estate (REIT):** VNQ, IYR, XLRE, REM, etc.
- **Commodities (Broad + Metals + Energy + Agriculture):** DBC, IAU, SLV, USO, CORN, WEAT, etc.

- **Commodity Miners / Materials:** GDX, LIT, COPX, URA, etc.
- **Thematic / Factor Suites:** AVUV, DFUS, JQUA, MOAT, etc.

For completeness, the full starting ticker set used in the data pull is:

IVV, VTI, ITOT, SCHB, DIA, QQQM, IJH, MDY, VO, IWR, IJR, VB, IWM, IWC, IWF, IWD, VUG, VTV, IWP, IWS, VOE, VBR, IWO, IWN, USMV, SPLV, MTUM, QUAL, SIZE, VLUE, RPV, RZG, RZV, VIG, VYM, SCHD, DVY, SDY, NOBL, HDV, SPYD, RSP, SCHA, SCHG, SCHV, SCHM, SCHX, VV, VONE, VTHR, XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY, XLC, XLRE, RYT, RYE, RYF, RGI, RTH, RYH, RHS, RYU, RCD, RTM, EWRE, SMH, SOXX, XSD, FDN, IGV, HACK, CIBR, SKYY, BOTZ, ROBO, IBB, XBI, IHE, IHI, ITA, XAR, IYT, IYJ, XME, SLX, XOP, OIH, ITB, XHB, XRT, IYR, VNQ, ICF, SCHH, KBE, KRE, IAI, IYF, IYZ, XTL, SOXQ, FTEC, VGT, XT, PAVE, ACWI, ACWX, VT, VEU, VEA, IEFA, EFA, VGK, IEV, VPL, EPP, VSS, SCZ, IEUS, DLS, EFV, IDEV, IQLT, EWJ, EWG, EWU, EWQ, EWC, EWA, EWL, EWH, EWS, EWP, EWD, EWN, EWI, EIRL, EWO, EWK, ENZL, PGAL, EDEN, EFNL, ENOR, GREK, EEM, VWO, IEMG, EWZ, EWT, EWY, EWW, EWM, EZA, FXI, MCHI, ASHR, KWEB, CQQQ, INDA, EPI, TUR, THD, EIDO, EPHE, VNM, EGPT, EIS, FM, KSA, QAT, UAE, KWT, PAK, ARG, GXG, GULF, GXC, KBA, NGE, EPU, ECH, DXJ, HEFA, HEDJ, BND, SCHZ, BNDW, GOVT, SGOV, SHY, SCHO, VGSH, SPTS, IEF, SPTI, VGIT, TLT, SPTL, VGLT, ZROZ, EDV, VGSH, VGIT, VGLT, LQD, IGSB, IGIB, VCIT, VCSH, HYG, JNK, USHY, SJNK, ANGL, HYL, PHB, HYS, SPHY, SPLB, SPSB, SPBO, TIP, SCHP, STIP, VTIP, MBB, MUB, TFI, VTEB, HYD, SHM, SMMU, SUB, CMF, NYF, TFSA, SHYD, BKL, SRLN, PFF, PFFD, PFXF, PFFA, PREF, PGX, PGF, VRP, IUSB, JPST, ICSH, NEAR, MINT, GSY, ULST, VNQ, IYR, REET, XLRE, RWR, VNQI, FREL, PPTY, NETL, HOMZ, RWO, REM, DBC, PDBC, COMT, BCI, CMDY, GSP, GCC, IAU, SLV, PPLT, PALL, OUNZ, GLTR, SGOL, USO, BNO, UNG, UGA, DBB, CPER, JJM, DBA, WEAT, CORN, SOYB, CANE, JO, NIB, SGG, COW, KRB, GRN, GDX, GDXJ, SIL, SILJ, COPX, LIT, REMX, URA, XME, SLX, ARKK, ARKG, ARKW, ICLN, TAN, PBW, FAN, PAVE, PHO, FIW, CLOU, WCLD, FIVG, IBUY, FINX, IPAY, JETS, BETZ, ESPO, HERO, UFO, MOAT, OMFL, PPA, QQQJ, QQQS, XT, BUG, VFMO, VFQY, VFMF, VFMV, AVUV, AVUS, AVL, AVDV, AVDE, AVEM, DFAU, DFUS, DFIV, DFHY, DFEM, DFAT, DFUV, JQUA, JMIN, JVAL, JHML, JPSE, JPRE, COWZ, CALF, BBJP, UUP, FXE, FXB, FXY, FXF, CEW, IAK, IHF, IHI, IYG, IYE, IYM, IYZ, IEO, IEZ, IETC, IUSG, IUSV, XTN, XAR, XHE, XHS, XHB, XPH, XNTK, XWEB, XRT, KBWB, KBWY, KBWR, PBJ, PEJ, PXJ, PXE, PBE, PSP, PKW, PSCT, PSCD, PSCM, PSCC, PSCE, PSCQ, OEF, SCHK, SCHP, SCHV, SCHX, SCHZ, SCZ, ESGV, SUSA, VGT, VHT, VIOO, VIOV, VO, VOE, VPL, VSS, VT, VTIP, VTI, VTV, VUG, VYM, VGIT, VGK, VGLT, VGSH, VNQ, VNM, VPL, VTEB, VRP, EUSA, FBT, IFRA, ILCG, ILCV, IXUS, IYW, MBB, NETL, NYF, PCEF, PSJ, QAT, RCD, ROBO, RTH, XSD, XT, XTL, XTN, ZROZ.

A.2. Survivorship and duplication handling

Multiple ETFs in the universe track nearly identical benchmarks (e.g., IVV / VOO / SPY for the S&P 500). To avoid artificially overweighting specific exposures, we remove exact or near-duplicate funds when

1. they track the same index with the same construction methodology, and
2. one fund is strictly dominated in age, assets, or liquidity.

This yields the **414 unique tickers** used at the start of the data-quality screen in Section 2.3.

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